

VEGUPATHI

Backend Engineer | Python • FastAPI • Distributed Systems • AI Voice Integrations

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PROFESSIONAL SUMMARY

Detail-oriented and execution-driven **Backend Engineer** with solid engineering fundamentals and hands-on expertise in **Python (FastAPI, Django)**, RESTful APIs, real-time streaming protocols (**WebSockets**), and database management (**PostgreSQL, Redis**). Proven capability in architecting low-latency microservices, integrating Large Language Model (LLM) voice workflows (Speech-to-Text, Text-to-Speech), and deploying containerized applications with **Docker and Cloud services**. Passionate about building highly reliable, production-ready systems for conversational AI and sales automation platforms.

CORE TECHNICAL SKILLS

Backend & Languages: Python (FastAPI, Django, Flask, Asyncio), JavaScript (ES6+), SQL, Bash scripting

Real-Time & Streaming: WebSockets, Asynchronous Event Loops, RESTful APIs, WebRTC concepts, Webhooks

AI & Voice Integration: OpenAI / Gemini APIs, Speech-to-Text (Whisper / STT), Text-to-Speech (TTS), Prompt Engineering, AI Agents

Databases & Queues: PostgreSQL, Redis (Caching & Pub/Sub), Celery Distributed Queues, MySQL, MongoDB

DevOps & Cloud: Docker, Kubernetes (Basics), AWS (EC2, S3), Git/GitHub, CI/CD Pipelines, Linux/Unix

FEATURED TECHNICAL PROJECTS

Real-Time Multilingual Conversational AI Agent Backend

FastAPI, WebSockets, Redis, OpenAI/Gemini API, Whisper STT

- Designed and implemented an asynchronous backend service handling bidirectional real-time audio and text streaming via **WebSockets** with sub-second response latency.
- Integrated Speech-to-Text (Whisper/STT) and low-latency Text-to-Speech (TTS) pipelines to enable natural, human-sounding voice conversations across Indian language contexts.
- Leveraged **Redis** for stateful session persistence and conversation context management, enabling robust multi-turn dialogues and dynamic objection handling.
- Engineered automated lead-routing logic to trigger real-time human handoffs when high-intent customer responses are detected.

Automated Lead Qualification & Campaign Orchestration API

Python, Django REST Framework, PostgreSQL, Celery, Redis, Docker

- Engineered a scalable backend microservice managing outbound customer contact campaigns, contact lists, and automated call dispatching.
- Utilized **Celery & Redis** for distributed asynchronous task execution, handling 10,000+ daily campaign records and telemetry updates without system bottlenecks.
- Developed robust webhook endpoints to consume telephony call event logs, track call outcomes, and generate structured client analytics reports.
- Containerized services using **Docker** to ensure 100% environment parity across staging and production deployments.

High-Performance Telephony & Event Observability Service

FastAPI, PostgreSQL, WebSockets, Docker, Prometheus / Logging

- Developed backend middleware to capture live call telemetry, SIP/call status events, and latency metrics across communication pipelines.
- Implemented structured logging, error alerting, and health check monitoring to diagnose and resolve production exceptions rapidly.

- Optimized database queries and PostgreSQL index strategies, reducing API endpoint response times by 35%.

PROFESSIONAL EXPERIENCE

Backend Developer / Freelance Software Engineer2023 – Present

Remote • Client Solutions & Backend Architecture

- Architected and delivered custom backend services, RESTful APIs, and full-stack web applications for commercial clients.
- Implemented secure token-based authentication (JWT, OAuth2) and role-based access control (RBAC) ensuring strict client data isolation.
- Resolved production exceptions, diagnosed concurrency bottlenecks, and optimized server configurations for high throughput.
- Maintained an execution mindset by translating business requirements into clean, testable, and maintainable Python code.

EDUCATION

Bachelor of Engineering / B.Tech / B.Sc in Computer Science / IT2020 – 2024

Anna University / Affiliated College, Tamil Nadu, India

KEY STRENGTHS

- **High Ownership:** Self-driven engineer who takes complete accountability for feature reliability from development to production monitoring.
- **Fast Learner:** Proven ability to quickly master new cloud stacks, real-time protocols (WebSockets/WebRTC), and AI voice frameworks.
- **Clear Communicator:** Seamlessly collaborates with product, operations, and technical teams to deliver practical customer solutions.